Project Part 6: Predefined queries with output

1. Calculate the total Fines by Member

```
SELECT
  bt.ClientID,
  c.Name                  AS MemberName,
  SUM(
    GREATEST(
      DATEDIFF(
        COALESCE(bt.ReturnDate, CURRENT_DATE),
        bt.DueDate
      ), 0
    )
    * m.DailyFineRate
  )                       AS TotalFine
FROM borrowtransaction bt
JOIN client     c ON bt.ClientID          = c.ClientID
JOIN membership m ON c.MembershipTypeID = m.MembershipTypeID
GROUP BY bt.ClientID, c.Name;
```

Output Screenshot:

Limit to 1000 rows

```sql
SELECT
  bt.ClientID,
  c.Name            AS MemberName,
  SUM(
    GREATEST(
      DATEDIFF(
        COALESCE(bt.ReturnDate, CURRENT_DATE),
        bt.DueDate
      ), 0
    )
    * m.DailyFineRate
  )                  AS TotalFine
FROM borrowtransaction bt
JOIN client      c ON bt.ClientID        = c.ClientID
JOIN membership m ON c.MembershipTypeID = m.MembershipTypeID
GROUP BY bt.ClientID, c.Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ClientID	MemberName	TotalFine
100708262468	Justin Roberts	0.00
106325369465	Amanda Zavala	0.00
158774376402	Stephanie Williams	0.00
242955608655	Robert Giles	0.00
386979565044	William Garcia	0.00
534771852898	David Moore	0.00
588022882666	Mrs. Meghan Stewart	0.00
609741931457	Michael Grant	0.00
777315144757	Eddie Martinez	0.00
862536141388	Stephanie Lawrence	0.00
926509076716	Amy Stark	0.00
108706256089	Brooke Mccarthy	0.50
136178644349	Steven Henderson	0.00
363714941482	Nicholas Kline	0.00
600450083709	Autumn Morse	11.25
44756014165	Timothy Garza	113.00
65543315395	Elizabeth Vincent	3.00
72458122216	Angel Campbell	45.00
80444014843	Monique Santana	0.00
98844556456	Antonio Mcdonald	155.00
151534339699	Elizabeth Houston	0.00
152491468918	Brian Cross	18.00
162788563211	Danielle Browning	0.00
345462137506	Melissa Moore	0.00
368798545706	Sandra Miller	41.00
392888992260	Christine Tran	455.00
664904820126	Alyssa Reid	0.00
670019806533	Thomas Harris	448.00
779102014641	Norma Fisher	460.00
908358000255	Autumn Robinson	439.00
963447298338	Chris Cook	0.00

2. Display a list of all available books

```
SELECT
    b.ISBN,
    b.Title,
    b.Author,
    b.Genre
FROM book b
JOIN copy cp ON b.CopyID = cp.CopyID
WHERE cp.AvailabilityStatus = 'Available'
    AND b.Genre = 'Mystery';
```

Output Screenshot:

```sql
SELECT
  b.ISBN,
  b.Title,
  b.Author,
  b.Genre
FROM book b
JOIN copy cp ON b.CopyID = cp.CopyID
WHERE cp.AvailabilityStatus = 'Available'
  AND b.Genre = 'Mystery';
```

Result Grid | Filter Rows: | Export: | Wra

ISBN	Title	Author	Genre
1023832279531	Twilight Covenant	Orion Black	Mystery
MST0001	Mystery Dawn	Alice Chen	Mystery
MST0002	Mystery Dusk	Bob Li	Mystery
MST0005	Mystery Echoes	Eve Zhang	Mystery
MST0006	Mystery Winds	Frank Gao	Mystery
MST0008	Mystery Codes	Henry Wu	Mystery

3. Identify the members who have borrowed the most books in a particular genre in last year

```
SELECT
    bt.ClientID,
    c.Name                    AS MemberName,
    COUNT(*)                  AS BorrowCount
FROM borrowtransaction bt
JOIN client c        ON bt.ClientID = c.ClientID
JOIN book     b        ON bt.CopyID     = b.CopyID
WHERE b.Genre = 'Mystery'             -- ← 类型
    AND bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1 YEAR)
GROUP BY bt.ClientID, c.Name
ORDER BY BorrowCount DESC
LIMIT 1;
```

Output Screenshot:

```sql
SELECT
  bt.ClientID,
  c.Name          AS MemberName,
  COUNT(*)        AS BorrowCount
FROM borrowtransaction bt
JOIN client c  ON bt.ClientID = c.ClientID
JOIN book    b  ON bt.CopyID    = b.CopyID
WHERE b.Genre = 'Mystery'
  AND bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1 YEAR)
GROUP BY bt.ClientID, c.Name
ORDER BY BorrowCount DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ClientID	MemberName	BorrowCount
100708262468	Justin Roberts	2

4. Generate a report of all books due within the next week, sorted by due date

```
SELECT
    b.ISBN,
    b.Title,
    bt.DueDate,
    c.Name AS Borrower
FROM borrowtransaction bt
JOIN client c ON bt.ClientID = c.ClientID
JOIN book     b ON bt.CopyID     = b.CopyID
WHERE bt.ReturnDate IS NULL
    AND bt.DueDate BETWEEN CURRENT_DATE
                                         AND DATE_ADD(CURRENT_DATE, INTERVAL 7
DAY)
ORDER BY bt.DueDate;
```

Output Screenshot:

```sql
SELECT
  b.ISBN,
  b.Title,
  bt.DueDate,
  c.Name AS Borrower
FROM borrowtransaction bt
JOIN client c ON bt.ClientID = c.ClientID
JOIN book   b ON bt.CopyID   = b.CopyID
WHERE bt.ReturnDate IS NULL
  AND bt.DueDate BETWEEN CURRENT_DATE
                     AND DATE_ADD(CURRENT_DATE, INTERVAL 7 DAY)
ORDER BY bt.DueDate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ISBN	Title	DueDate	Borrower
8049194392752	The Forgotten Kingdom	2025-05-07	Antonio Mcdonald
8049194392752	The Forgotten Kingdom	2025-05-08	Justin Roberts
9158736461602	Dreams of Glass	2025-05-08	Elizabeth Vincent

5. List all members who currently have at least one overdue book, along with the titles of the overdue books.

```
SELECT
    bt.ClientID,
    c.Name                          AS MemberName,
    GROUP_CONCAT(b.Title SEPARATOR ', ') AS OverdueTitles
FROM borrowtransaction bt
JOIN client c       ON bt.ClientID = c.ClientID
JOIN book      b       ON bt.CopyID      = b.CopyID
WHERE bt.ReturnDate IS NULL
    AND bt.DueDate < CURRENT_DATE
GROUP BY bt.ClientID, c.Name;
```

Output Screenshot:

```sql
SELECT
  bt.ClientID,
  c.Name               AS MemberName,
  GROUP_CONCAT(b.Title SEPARATOR ', ') AS OverdueTitles
FROM borrowtransaction bt
JOIN client c    ON bt.ClientID = c.ClientID
JOIN book   b    ON bt.CopyID   = b.CopyID
WHERE bt.ReturnDate IS NULL
  AND bt.DueDate < CURRENT_DATE
GROUP BY bt.ClientID, c.Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ClientID	MemberName	OverdueTitles
72458122216	Angel Campbell	Echoes of Tomorrow
98844556456	Antonio Mcdonald	The Crystal Code, Dreams of Glass
242955608655	Robert Giles	Shadows of Eternity
368798545706	Sandra Miller	The Forgotten Kingdom
600450083709	Autumn Morse	Whispers in the Wind
779102014641	Norma Fisher	Beneath the Iron Sky

6. Calculate the average number of days members borrow books for a specific genre.

```
SELECT
    b.Genre,
    AVG(
        DATEDIFF(
            COALESCE(bt.ReturnDate, CURRENT_DATE),
            bt.CheckoutDate
        )
    ) AS AvgBorrowDays
FROM borrowtransaction bt
JOIN book b          ON bt.CopyID = b.CopyID
WHERE b.Genre = 'Mystery'
GROUP BY b.Genre;
```

Output Screenshot:

```sql
SELECT
  b.Genre,
  AVG(
    DATEDIFF(
      COALESCE(bt.ReturnDate, CURRENT_DATE),
      bt.CheckoutDate
    )
  ) AS AvgBorrowDays
FROM borrowtransaction bt
JOIN book b      ON bt.CopyID = b.CopyID
WHERE b.Genre = 'Mystery'
GROUP BY b.Genre;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Genre	AvgBorrowDays
Mystery	16.4000

7. Determine the author whose books have been borrowed the most in the last month.

```
SELECT
    b.Author,
    COUNT(*)                AS BorrowCount
FROM borrowtransaction bt
JOIN book b             ON bt.CopyID = b.CopyID
WHERE bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1
MONTH)
GROUP BY b.Author
ORDER BY BorrowCount DESC
LIMIT 1;
```

Output Screenshot:

```sql
SELECT
  b.Author,
  COUNT(*)          AS BorrowCount
FROM borrowtransaction bt
JOIN book b         ON bt.CopyID = b.CopyID
WHERE bt.CheckoutDate >= DATE_SUB(CURRENT_DATE, INTERVAL 1 MONTH)
GROUP BY b.Author
ORDER BY BorrowCount DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Author	BorrowCount
Damien Hale	3